BOILEAU GUILLAUME

Ingénieur Validation Vérification Logiciel - IVVQ | Systèmes Critiques, Tests, Intégration & Exigences

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EXPERIENCE

Software Engineer - Integration, Verification, Validation & Production Follow-up

VEV Platform Services France

2025 – 2026 | Valbonne, France

Context: Development and integration of software services for an EV charging CPMS platform connected to operational infrastructure and field equipment.
Objective: Support reliable delivery, production follow-up, software integration and operational data-flow continuity between platform services and charging station systems.

- **Software delivery:** Developed and maintained web and backend services using TypeScript, Angular and Node.js in an operational industrial environment.
- **System integration:** Contributed to the integration and maintenance of software services interacting with field equipment, charging stations and operational infrastructure.
- **Needs clarification:** Translated operational issues, data-flow constraints, equipment feedback and user needs into technical tasks and corrective actions.
- **Production follow-up:** Analysed service behaviour, data exchanges, communication issues and anomalies affecting operational continuity.
- **Verification / validation:** Checked data consistency, interface continuity and service behaviour in production-like or operational conditions.
- **Technical issue management:** Investigated incidents involving software behaviour, operational data exchanges and hardware-connected systems.
- **Coordination:** Worked with technical stakeholders to follow priorities, clarify issues and support iterative delivery.
- **Data exchange:** Implemented and monitored FTP-based operational data exchange services.
- **Field / equipment exposure:** Hands-on exposure to EV charging stations, test support and electrical-safety constraints in an industrial context.
- **Agile tooling:** Used Zenhub to support task tracking, backlog follow-up, iterative development and technical coordination.

Senior R&D Engineer - Software IVVQ, Validation & Technical Coordination

CNES, Laboratoire Lagrange, Observatoire de la Côte d'Azur

2023 – 2025 | Nice, France

Context: Engineering activities for the LISA ESA/NASA space mission, involving complex software chains, model integration, simulation workflows, validation campaigns and international coordination.
Objective: Structure, coordinate and validate complex software/data workflows under strong consistency, traceability, performance and verification constraints.

- **Project coordination:** Coordinated technical activities involving contributors, research engineers, technicians and Master's thesis interns.
- **Specifications / cadrage:** Structured development priorities, validation logic, expected outputs and technical acceptance criteria for complex analysis workflows.
- **Solution implementation:** Developed CATpyLISA, a Python platform for large-scale model integration, comparison, validation and technical review. GitLab: CATpyLISA
- **V-cycle logic:** Supported activities aligned with specification, integration, verification, validation, traceability and technical evidence consolidation.
- **Verification / validation:** Defined comparison campaigns, consistency checks, acceptance logic and validation workflows for technical outputs.
- **Risk follow-up:** Identified deviations, inconsistencies, limiting factors and technical risks affecting result reliability or delivery quality.
- **Reporting:** Produced technical notes, validation summaries, analysis reports and review material for contributors and project stakeholders.
- **Documentation:** Used Confluence to support technical documentation, coordination notes, validation follow-up and knowledge sharing.
- **Stakeholder alignment:** Worked at the interface between scientific requirements, software development, data processing, modelling assumptions and mission-level objectives.
- **International environment:** Operated in a collaborative framework requiring technical English, structured communication and versioned workflows.

Data Scientist - Technical Studies, Performance Analysis & Project Reporting

Universiteit Antwerpen, Belgium

2021 – 2023 | Antwerp, Belgium

Context: Data analysis and performance studies for precision instruments in a high-requirement international environment linked to gravitational-wave detector performance.
Objective: Identify, characterize and mitigate factors affecting system sensitivity using statistical, computational and validation-oriented methods.

- **Technical studies:** Analysed instrumental and environmental effects impacting system sensitivity, stability and measurement quality.
- **Analysis workflows:** Developed Python-based pipelines for noise source identification, uncertainty estimation and statistical modelling.

PROFILE FOR THALES IVVQ

Software IVVQ / systems engineer with strong experience in requirements analysis, test strategy, integration, verification, validation, technical documentation and delivery support for complex and mission-critical systems. Background developed in international space-mission environments and reinforced by recent industrial software experience involving operational services, hardware/software interfaces and production follow-up.
For a **Software Verification, Validation and Integration** role, I bring a strong V-cycle and delivery-oriented profile: analysing requirements, preparing validation logic, defining and executing test campaigns, analysing results, documenting evidence, supporting integration activities and communicating clearly with clients and stakeholders.

- Strong fit with software IVVQ activities: requirements analysis, test preparation, verification, validation, result analysis, documentation and stakeholder coordination.
- Experience coordinating multidisciplinary contributors across software, data, modelling, operational constraints and system-level objectives in critical technical environments.
- Ability to translate requirements, anomalies and integration constraints into structured test actions, corrective follow-up and delivery evidence.
- Strong analytical ability to investigate technical issues, analyse deviations, identify risks, challenge assumptions and converge toward pragmatic solutions.
- Practical experience with software integration, APIs, data flows, Linux environments, containers, Git-based workflows and operational monitoring.
- Industrial exposure to hardware/software interfaces through EV charging infrastructure, operational data exchanges, equipment-connected services and test support.
- Comfortable working in English and in multidisciplinary environments involving technical and non-technical stakeholders.

FIT FOR SOFTWARE IVVQ

- Relevant background for IVVQ phases: requirements analysis, test strategy support, integration follow-up, verification, validation, documentation and delivery support.
- Strong experience producing test-oriented documentation, validation reports, analysis summaries, technical notes and decision-support material.
- Experience preparing validation campaigns, coordinating contributors, clarifying priorities, identifying risks and supporting corrective actions.
- Practical understanding of software components: APIs, backend services, data flows, containers, Linux, Git/GitLab, CI/CD and operational monitoring.
- Experience at the interface between software behaviour, operational data, system constraints, hardware-connected infrastructure and user/client expectations.
- Comfortable with collaborative tools including Git/GitLab, GitHub, Confluence, Zenhub and JIRA-style issue tracking.

LANGUAGES

English ●●●●○

Spanish ●●○○○

EDUCATION

PhD in Physics

Université Côte d'Azur

- **Performance analysis:** Identified contributors to performance degradation and supported prioritisation of mitigation directions.
- **Data-driven decisions:** Analysed measurement and simulation outputs to identify abnormal behaviours and dominant contributors.
- **Documentation:** Produced reproducible and traceable technical analyses for international engineering and research stakeholders.
- **Coordination:** Communicated complex technical results through structured reporting, presentations and collaborative review.

Junior R&D Engineer - Simulation, Software Workflows & Validation

ARTEMIS, Observatoire de la Côte d'Azur

2018 – 2021 | Nice, France

Context: Engineering and analysis activities for gravitational-wave mission preparation, involving signal analysis, simulation, software workflows and noise characterization.
Objective: Develop robust analysis methods and validation workflows for complex signals and demanding system environments.

- **Simulation workflows:** Built simulation approaches for complex signal environments, uncertainty studies and large-scale datasets.
- **Software validation:** Compared simulated outputs, model behaviours and analysis results to assess consistency and robustness.
- **Technical investigations:** Investigated noise sources through cross-sensor correlation analyses in diagnostic contexts.
- **Performance analysis:** Supported the identification of limiting factors affecting mission-level performance and system sensitivity.
- **Scientific computing:** Developed strong expertise in Python-based analysis, modelling, documentation and reproducible workflows.

2018 – 2021 | Nice, France

Selective Graduate Program in Fundamental Physics (Magistère)

Université Paris-Saclay

2016 – 2018 | Orsay, France

MSc in Astronomy and Astrophysics

Observatoire de Paris / Université Paris-Saclay

2017 – 2018 | Paris, France

BSc in Fundamental Physics

Université Paris-Saclay

2015 – 2016 | Orsay, France

PROFESSIONAL TRAINING

Supervised Machine Learning (Regression & Classification)

DeepLearning.AI - Andrew Ng

2020 | Coursera

Recherche reproductible : principes méthodologiques pour une science transparente

FUN MOOC – Inria

2025 | France Université Numérique

Machine Learning in Python with scikit-learn

FUN MOOC – Inria

2026 | France Université Numérique

Continuous Professional Training – Software, Data, AI and Systems

OpenClassrooms

2024 – 2026

Python, SQL, Machine Learning, AI, Linux, JavaScript, TypeScript, Angular, Node.js, Django, REST APIs, C/C++, web development, algorithms and software engineering fundamentals.

PROJECTS

CNES / LISA – Software IVVQ, Requirements Logic & Validation Evidence

ESA/NASA Space Mission - Large-scale International Collaboration

2023 – 2025 | Nice, France

Coordination and structuring of software, data-processing and validation workflows for the LISA space mission within a large international engineering and scientific collaboration.
Project management relevance: Cadrage of technical activities, contributor coordination, validation planning, delivery follow-up, risk identification, reporting, documentation and stakeholder alignment.
IVVQ relevance: Work involving complex software workflows, data-processing chains, GitLab-based development, reproducible environments, technical documentation, validation evidence and versioned deliverables.
Scale: Major international space mission with an estimated budget of approximately 2 billion EUR over 10 years.

Einstein Telescope / LIGO–Virgo–KAGRA – Technical Studies & Collaborative Project Workflows

International Collaborations

2021 – 2023 | Antwerp, Belgium

Contribution to collaborative developments in data analysis, software methods and technical investigations within large-scale international scientific and engineering collaborations.
Project management relevance: Coordination of technical studies, reporting, documentation, contribution to collaborative review, prioritisation of analysis tasks and communication with international stakeholders.
System impact: Studies on environmental and instrumental noise sources supporting the identification of performance limitations and design constraints for next-generation gravitational-wave observatories.

STRENGTHS

IVVQ, delivery & client satisfaction

IVVQ · Integration · Verification · Validation · Test Strategy · Test Campaigns · Manual Tests · Automated Tests Awareness · Requirements Analysis · Requirements Trace
Test Procedures · Result Analysis · Anomaly Investigation · Delivery Support · Client Demonstration · Quality Assurance · Risk Identification · Continuous Improvement

Critical software, test tools & environments

Critical Systems · Complex Software · System Integration · Hardware / Software Interfaces · Software Workflows · Test Environments · Regression Tests · Simulation Work
Data Consistency Checks · Acceptance Logic · Validation Evidence · Test Reports · Deployment Support · Production Follow-Up · User-Focused Problem Solving

Tools, languages & technical environment

Python · C / C++ Basics · Linux · Docker · CI/CD · Git / GitLab · GitHub · SQL · TypeScript · Angular · Node.js · OpenSearch · DOORS Awareness
JIRA · Confluence · Zenhub · Operational Monitoring · EV Charging Infrastructure

Methods, collaboration & documentation

Agile · Scrum Awareness · V-cycle · Integration Planning · Test Planning · Client Expectations · Stakeholder Influence · Collaborative Work · Cross-Functional Coordinatio
Technical Documentation · Procedure Writing · Review Material · Progress Reporting · Technical English

Office & communication

PowerPoint · Excel · Word · MS Office · Executive Presentations · Reporting Dashboards · Indicator Follow-Up · Synthesis · Analytical Thinking · Problem Solving
Pedagogy · Autonomy · Adaptability · Rigor